

Lecture 3: Preferences and Utility, Part II

Alejandro Melo Ponce



Stony Brook University

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Preferences and Utility

Definition

Preferences on a set X is a relation $\succsim$ on X (a way to relate elements of X) satisfying:

- **Completeness:** For any $x, y \in X$ either $x \succsim y$ or $y \succsim x$.
 - **Transitivity:** For any $x, y, z \in X$ such that $x \succsim y, y \succsim z$ then $x \succsim z$.
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- When a relation $\succsim$ satisfy completeness and transitivity, we call it a **rational preference relation**.
 - This term is misleading, so I prefer to say that a preference that satisfy completeness and transitivity is internally consistent.

Definition

A **utility function (index)** $u : X \rightarrow \mathbb{R}$ represents a preference relation $\succsim$ over X if and only if, for all $x, y \in X$ we have that $x \succsim y \Leftrightarrow u(x) \geq u(y)$.

When can we find such u ?: X is a finite set

Proposition

Suppose X is finite. Then the assumptions of completeness and transitivity imply that there exist a utility function u that represents the preference relation $\succsim$ on X

- Define the set $B(x) = \{z \in X \mid x \succsim z\}$.
- Let $u(x) = \#(B(x))$, i.e. the number of items on $B(x)$
- Proof:
 - First part: Showing that $u(x) \geq u(y) \Rightarrow x \succsim y$.
 - Second part: $x \succsim y \Rightarrow u(x) \geq u(y)$

What happens when X is infinite

- We need a technical definition to handle this case:

Definition

Let x_n and y_n be two sequences of elements of X . If $x_n \succsim y_n$ for every n and $x_n \rightarrow x$ and $y_n \rightarrow y$, then $x \succsim y$

- So we have in general:

Proposition

Every continuous preference relation can be represented by a continuous utility function.

- The proof of this result is harder and we will not cover it here.

Utility is a purely ordinal concept!

- The actual number does not matter, only the order it represents.
- Formally, if u represents preferences, then for any increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(u(x))$ also represents the same preferences.
- An increasing function is called **monotone**. $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic or increasing if and only if for every $x, y \in \mathbb{R}$ such that $x \geq y$, implies $f(x) \geq f(y)$.
- So what this result is saying is that any index that respects the order on X induced by the preference $\succsim$ works as a utility function.
- Rephrasing, if u represents $\succsim$, so does any monotonic transformation of it.

Representing Preferences: Indifference curves

- From now on, we will assume that our set $X = \mathbb{R}^n$, which we will interpret as the set of **consumption bundles**.
- An indifference curve is the set of points that are indifferent between each other:

$$I(x) = \{z \in X | z \sim x\}$$

- If preferences are represented by a utility function, we can express the same idea as follows:

$$I(x) = \{z \in X | u(z) = u(x)\}$$

- A graphical representation of an individual's preferences is by drawing their indifference curve map.

Indifference curve map

Each indifference curve represents a different utility level.

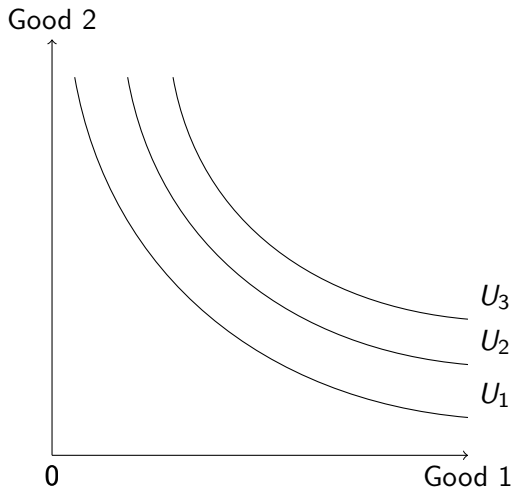
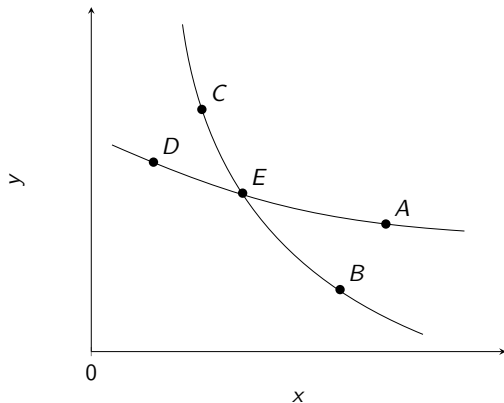


Figure: An indifference map

Transitivity and indifference curves

If preferences are transitive, indifference curves do not cross!



Suppose $A \succ B$ and $C \succ D$. From the picture, $A \sim D \sim E$ and $B \sim C \sim E$. Then by transitivity $A \sim B$ and $C \sim D$, which is a contradiction.

Homework: Try assuming now that $B \succ A$ and $D \succ C$.

Properties of preferences

Definition

Preferences $\succsim$ over $X = \mathbb{R}^n$ are said to be **monotone** (resp. **strictly monotone**) if $x \geq y$ implies $x \succsim y$ (resp. $x \geq y, x \neq y \Rightarrow x \succ y$.)

Remark

$x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are vectors in $\mathbb{R}^n$. We say that $x \geq y$ if and only if the inequality holds component-wise, i.e. $x_i \geq y_i$ for all i . $x \neq y$ means $x_i \neq y_i$ for at least one i .

Definition

A preference relation is convex if for all x and y with $x \sim y$ and for all $\alpha \in [0, 1]$ we have that $\alpha x + (1 - \alpha)y \succsim x \sim y$.

All of these properties can be stated in terms of Utility functions. More on this later...

Convexity and balance in consumption

- Intuitively, convexity has to do with an individual preferring more balanced consumption bundles than more extreme ones.
- Suppose you have bundles $(x_1, x_2) \sim (y_1, y_2)$. If preferences are convex, then the bundle $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ is preferred.
- This is usually assumed when we are talking about consumption bundles that are “economic goods”.
- However, convexity might not hold if the consumption bundle includes an item that might be considered a “bad”, like pollution.

Indifference curves and the marginal rate of substitution

- Consider the case $X = \mathbb{R}^2$. and assume that preferences are represented by u .
- Indifference curve U_1 is $\{(x, y) | u(x, y) = U_1\}$
- The equation $U_1 = u(x, y)$ defines, *implicitly* y as a function of x .
- If u is differentiable, we can think about calculating what is the slope of the indifference curve.
- This slope has a very important economic meaning:

Definition

The **marginal rate of substitution** is the *negative* of slope of an indifference curve U_1 at some point. That is

$$MRS = -\frac{dy}{dx} \Big|_{u=U_1} \quad (1)$$

the MRS tells us about the trades that an individual is willing to make, it is the opportunity cost in terms of y of consuming one unit more of x